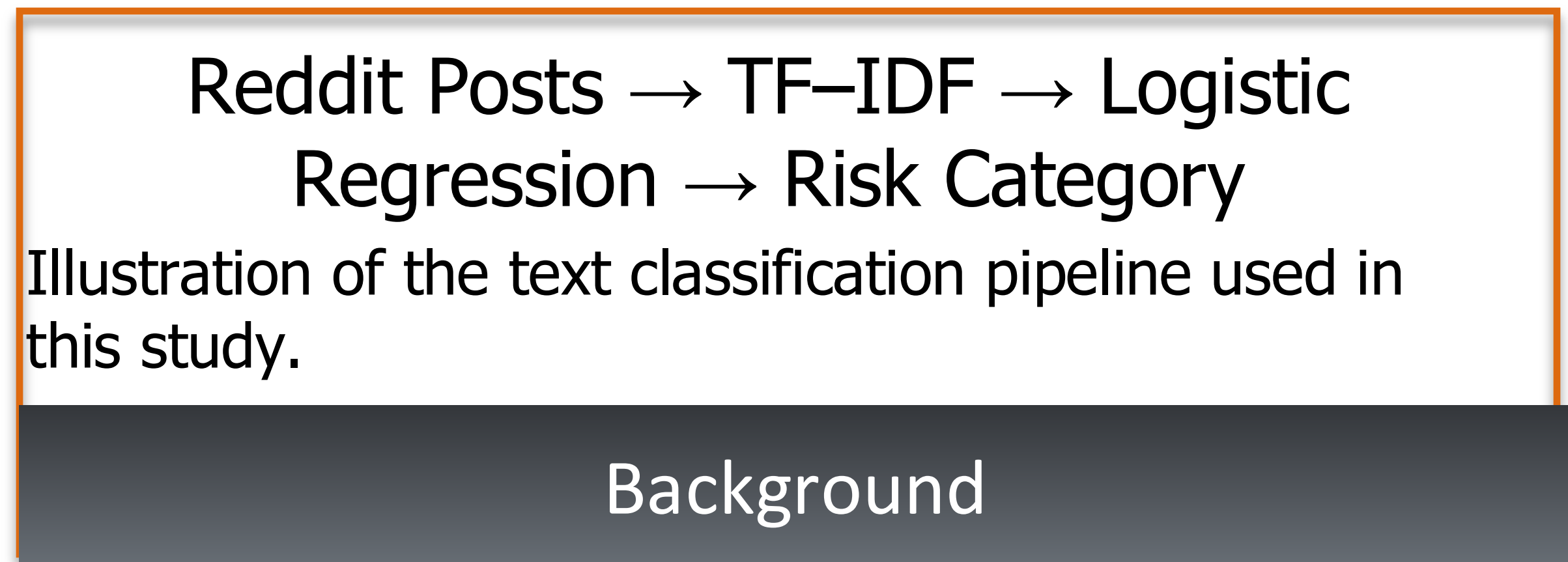


Motivation for your work

Data

Method

- Online posts can reveal early signs of suicidal thoughts, but signals are often subtle.
 - Moderators and crisis teams cannot review everything on large platforms like Reddit.
 - Identifying linguistic cues across risk levels helps improve:
 - Early crisis detection
 - Prioritizing high-risk posts
 - Safer online environments
 - Supports public-health research on how people express distress in real contexts.
 - Helps inform the development of better mental-health monitoring tools and future AI systems that can flag users needing urgent support.
- 500 anonymized Reddit mental-health posts (one per unique user)
 - Labeled into four categories: Supportive, Ideation, Behavior, Attempt
 - Text is short-medium length, informal, emotional, and conversational
 - Contains real expressions of distress, coping, and suicide-related communication
 - Supervised dataset with some class imbalance
 - Minimal cleaning needed before analysis (raw text with Reddit slang + punctuation)
 - Suitable for studying linguistic cues across different suicide-risk levels
- The project uses the provided TF-IDF-based text classification approach.
 - Posts are converted into TF-IDF feature vectors
 - A multiclass Logistic Regression model is trained to classify posts into:
 - Supportive, Ideation, Behavior, Attempt.
 - The dataset is preprocessed (cleaning, tokenizing, splitting into train/validation/test).
 - Evaluation uses accuracy, precision, recall, F1, and confusion matrices.



- Prior work shows that suicide risk can be inferred from language in Reddit posts.
- The CLPsych 2019 shared task focused on multi-level suicide risk classification, not just binary labels.
- Studies report that Supportive and Ideation posts are easier to classify than Behavior or Attempt.
- Overlap between ideation and behavior language is common and causes model confusion.
- TF-IDF with linear models is frequently used as a strong, interpretable baseline.

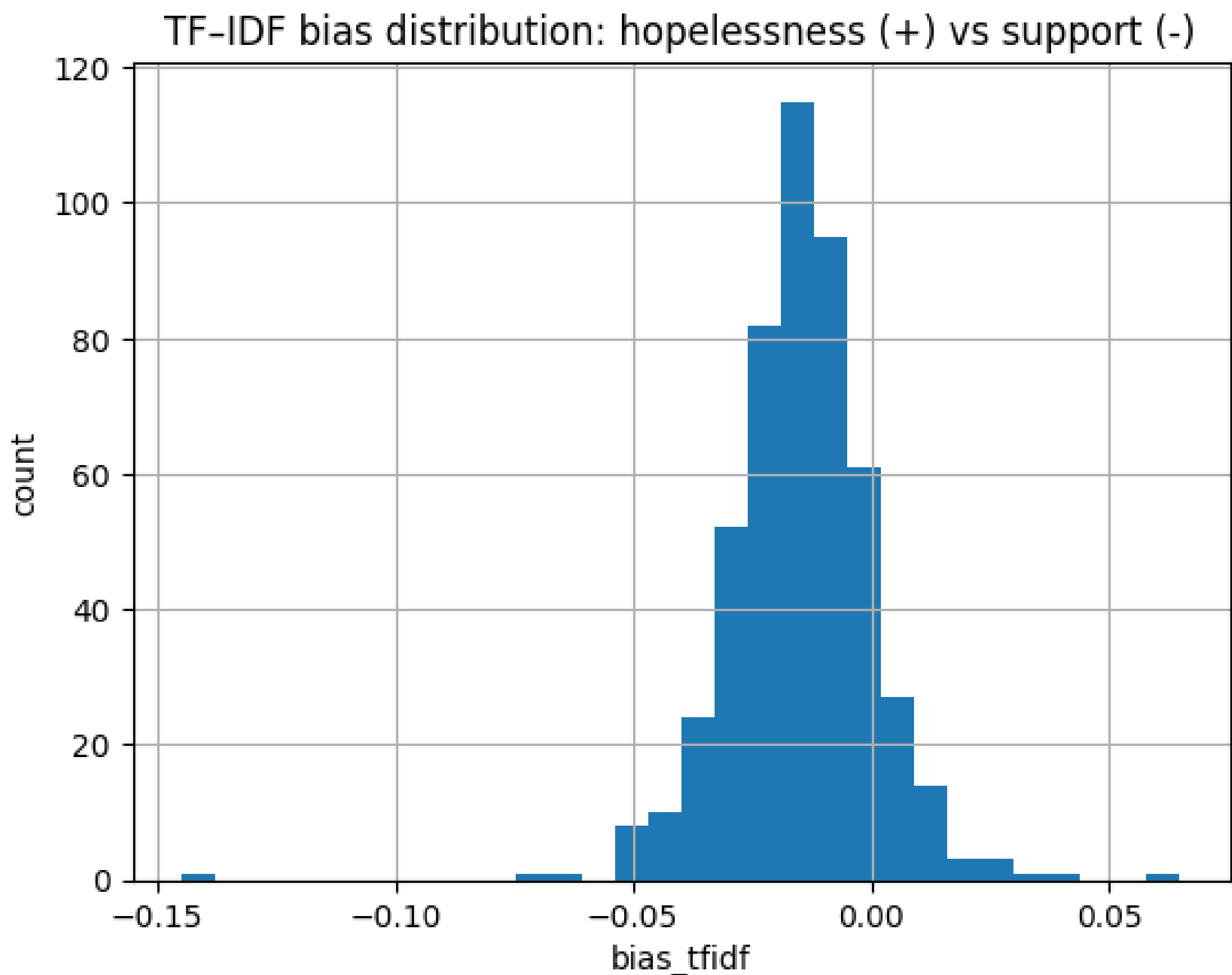
➤ **Language alone can reveal different suicide-risk levels** in Reddit posts.

➤ **TF-IDF is a strong, interpretable baseline** for mental-health text classification.

➤ **Supportive and Ideation posts are easier to identify** than Behavior and Attempt.

➤ **Class imbalance and language ambiguity matter** for model performance.

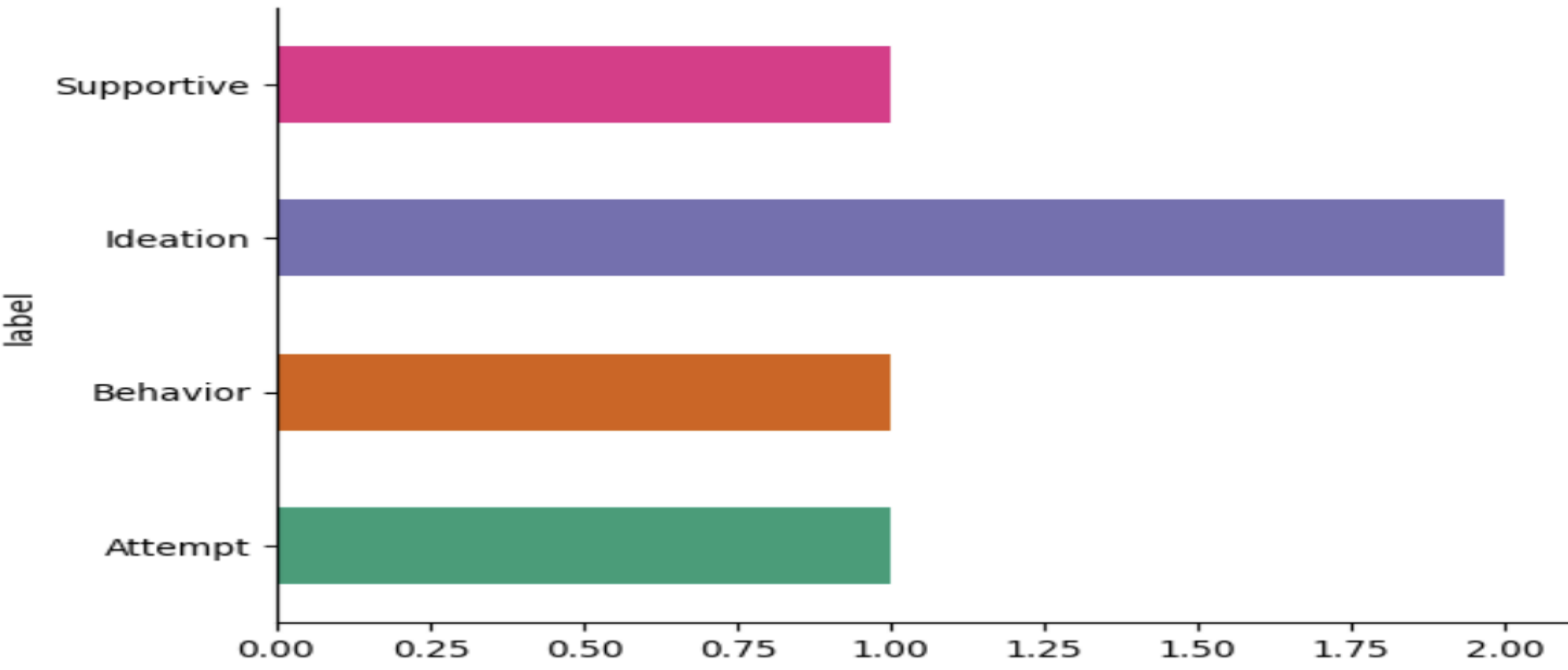
➤ Understanding these patterns can **support safer online spaces and early intervention**.



Results: quantify your results

Results: Quantify or explain

- Ideation and Supportive posts are more frequent than Behavior and Attempt, creating class imbalance that affects model performance.



The model most often confuses Ideation and Behavior, reflecting real-world overlap between suicidal thoughts and actions.

Confusion Matrix (rows=true, cols=pred):

	pred_Attempt	pred_Behavior	pred_Ideation	pred_Indicator	\
true_Attempt	0	3	3	2	
true_Behavior	1	5	6	1	
true_Ideation	0	6	11	9	
true_Supportive	0	0	5	8	

Attempt: attempt, somebody, suicide attempt, failed
Behavior: cutting, hospital, felt, kill im
Ideation: thank, depression, mental depression, mental
Supportive: think, sounds, therapist help, chocolate

- TF-IDF highlights meaningful words and phrases that distinguish different suicide-risk categories.